Mean, Median and Mode

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1. Mean of Daily Step Counts

Suppose the daily step counts of a person for 5 days are:

5,000, 7,000, 6,000, 8,000, 9,000

Formula:

Mean = Sum of all observations ÷ Number of observations

Calculation:

Mean = (5,000 + 7,000 + 6,000 + 8,000 + 9,000) ÷ 5

Mean = 35,000 ÷ 5

Mean = 7,000 steps

Conclusion: The average daily step count is **7,000 steps**.

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2. Median of Zomato Order Values

Given order values:

₹250, ₹400, ₹150, ₹600, ₹300

Step 1: Arrange the values in ascending order:

₹150, ₹250, ₹300, ₹400, ₹600

Step 2: Find the middle value.

There are 5 observations, so the middle value is the **3rd value**.

Median = ₹300

Conclusion: The median order value is **₹300**.

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3. Mode of Spotify Song Durations

Suppose a user's song durations are:

180, 240, 180, 300, 240, 180, 360 seconds

Frequency of each duration:

Song Duration	Frequency
180 seconds	3
240 seconds	2
300 seconds	1
360 seconds	1

The value occurring most frequently is **180 seconds**.

Mode = 180 seconds

Conclusion: The most frequently played song duration is **180 seconds**.

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4. Mean, Median and Mode of Instagram Likes

Array 1

Likes: 100, 120, 110, 130, 140

Mean:

Mean = (100 + 120 + 110 + 130 + 140) ÷ 5

Mean = 120

Median:

Arranged data:

100, 110, 120, 130, 140

Median = 120

Mode:

No value is repeated.

Mode = No mode

Best Measure: Mean or Median

Reason: The values are quite close to each other, so both mean and median represent a typical Instagram post well.

Array 2

Likes: 100, 110, 120, 130, 1000

Mean:

Mean = (100 + 110 + 120 + 130 + 1000) ÷ 5

Mean = 292

Median:

Arranged data:

100, 110, 120, 130, 1000

Median = 120

Mode:

No value is repeated.

Mode = No mode

Best Measure: Median

Reason: 1,000 likes is an unusually high value and increases the mean. Therefore, the median of **120** better represents a typical post.

Array 3

Likes: 200, 250, 200, 300, 200, 350

Mean:

Mean = (200 + 250 + 200 + 300 + 200 + 350) ÷ 6

Mean = 250

Median:

Arranged data:

200, 200, 200, 250, 300, 350

Since there are 6 observations:

Median = (3rd value + 4th value) ÷ 2

Median = (200 + 250) ÷ 2

Median = 225

Mode:

200 occurs **3 times**, which is the highest frequency.

Mode = 200

Best Measure: Mode

Reason: 200 likes occurs most frequently, so it represents the most common performance of a post.

Final Summary

Instagram Dataset	Mean	Median	Mode	Best Measure
Array 1	120	120	No Mode	Mean / Median
Array 2	292	120	No Mode	Median
Array 3	250	225	200	Mode